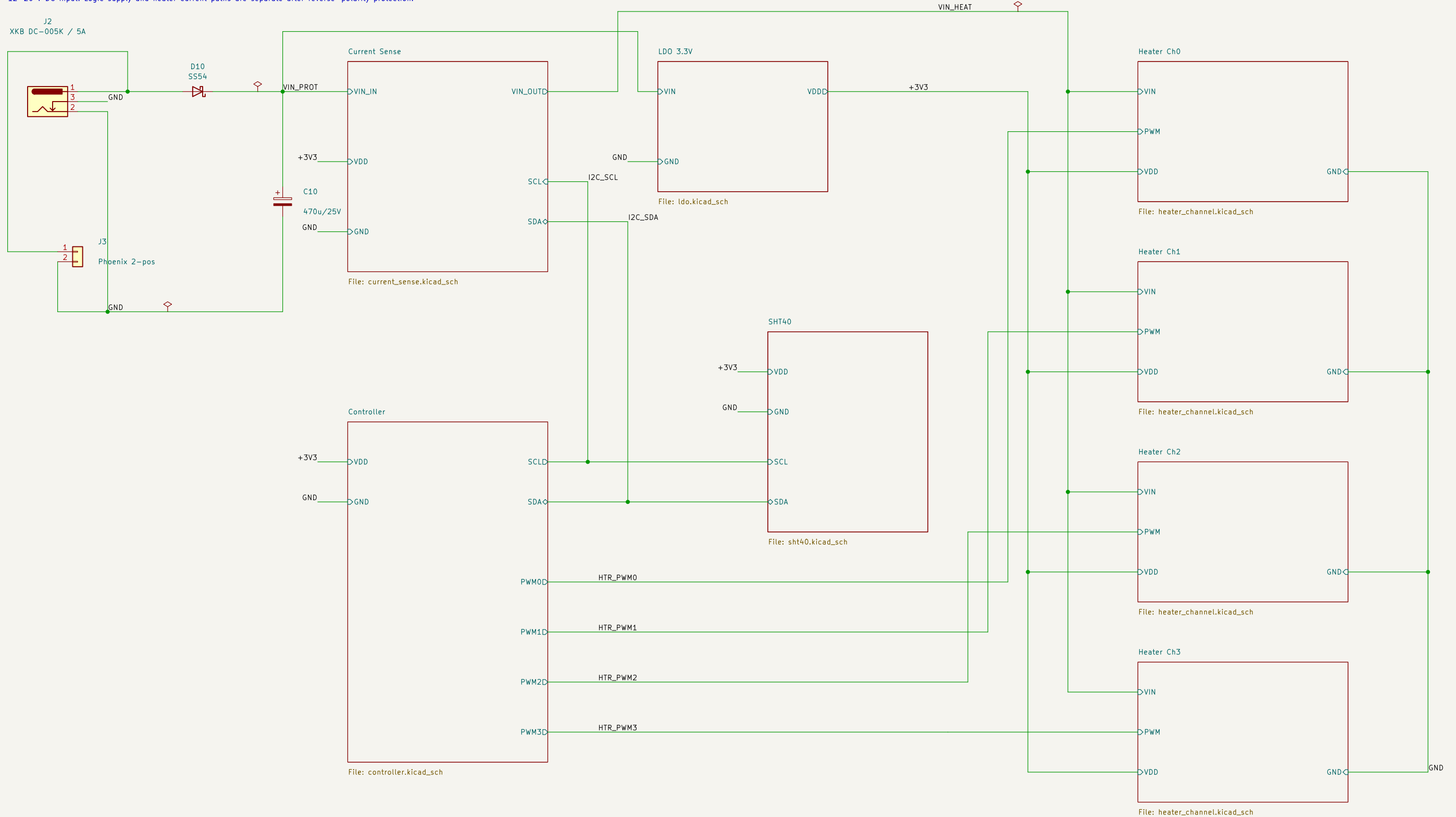


PULSARDEW | USB dew-heater controller

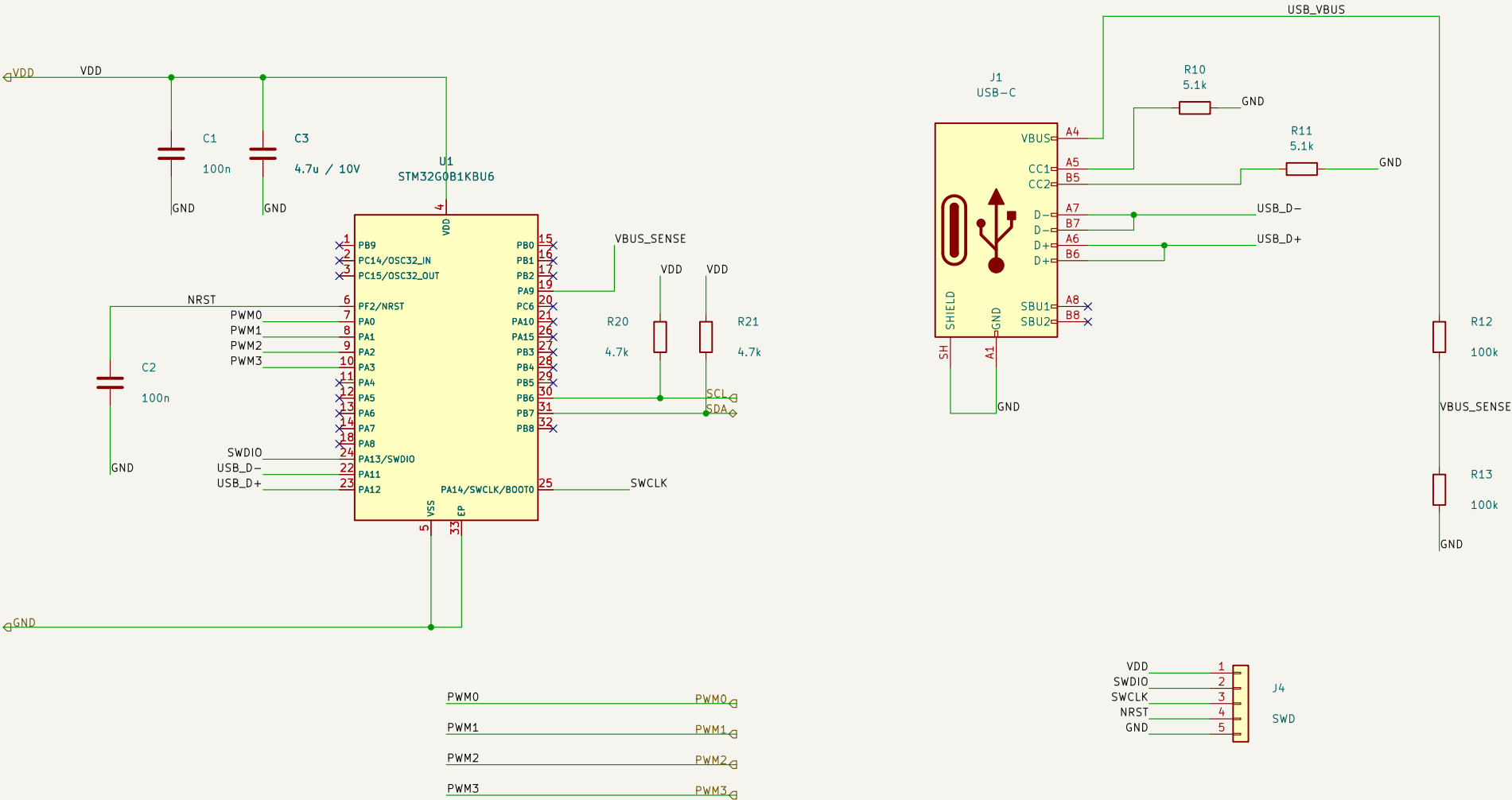
12–20 V DC input. Logic supply and heater current paths are separate after reverse-polarity protection.



Supply: J2 or J3 (one source). D10 protects polarity.
VIN_HEAT feeds all four heater connectors through R30.
One INA226 measures total heater current, excluding the logic LDO.
Current rating is limited by connectors, diode, shunt and PCB thermals.

CHANNELS
0: J101 / Q102 / PA0
1: J201 / Q202 / PA1
2: J301 / Q302 / PA2
3: J401 / Q402 / PA3

CONTROLLER | STM32G0B1KBU6, USB 2.0 FS, SWD



Self-powered from DC input. USB VBUS is sense-only (PA9); never join it to VIN or 3V3.
Firmware: attach USB only when VBUS is present; PA0..PA3 = TIM2 CH1..CH4, inverted duty.
PB6/PB7 = I2C1 SCL/SDA. PA13/PA14 = SWD. Keep default boot option bytes for flash boot.

PulsarFab

Sheet: /Controller/
File: controller.kicad_sch

Title: PulsarDew – MCU, USB and debug

Size: A4

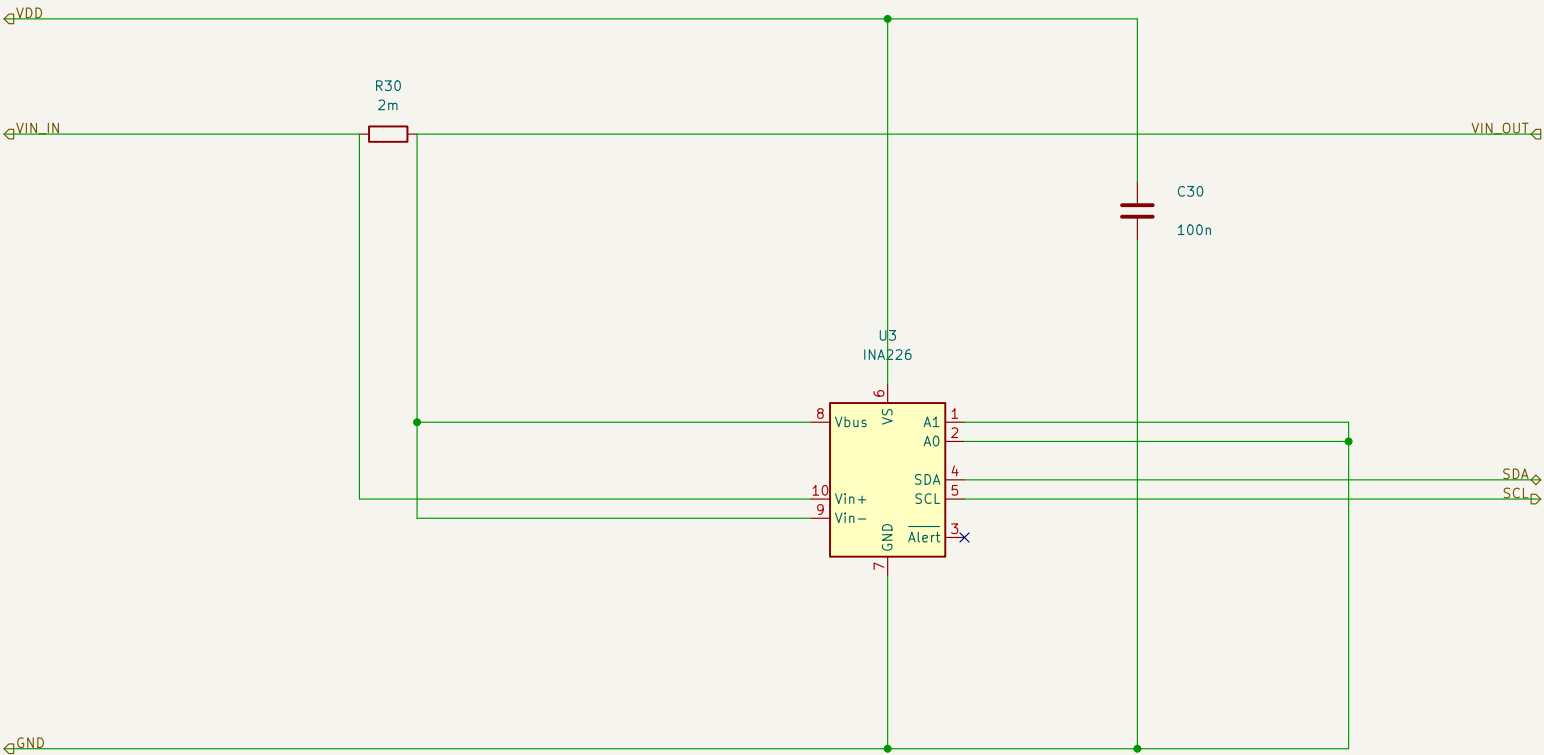
Date: 2026-10-02

Rev: 0.2

KiCad E.D.A. 10.0.6

Id: 2/9

CURRENT MONITOR | INA226, I2C address 0x40



Route IN+ / IN- as Kelvin pairs directly to the shunt pads. VBUS senses the load side.
A0 = A1 = GND. ALERT is unused. One monitor measures the combined heater current.
2 mOhm: 2.5 uV/LSB = 1.25 mA/LSB. ADC full scale is not the board current rating.

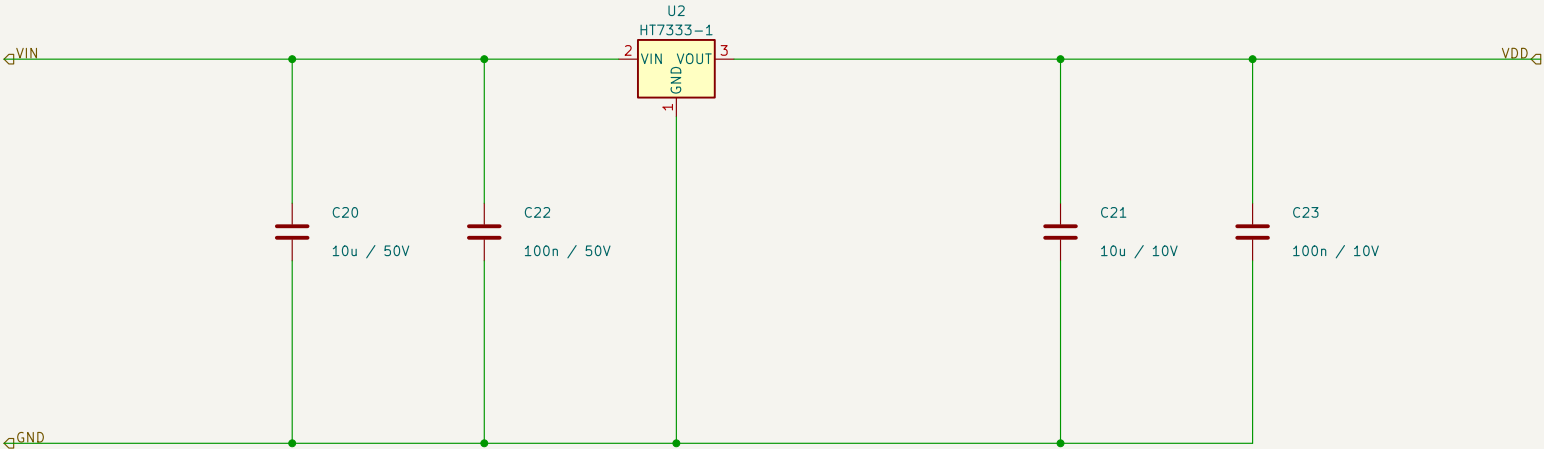
PulsarFab

Sheet: /Current Sense/
File: current_sense.kicad_sch

Title: PulsarDew – Input current monitor

Size: A4	Date: 2026-10-02	Rev: 0.2
KiCad E.D.A. 10.0.6	Id: 3/9	

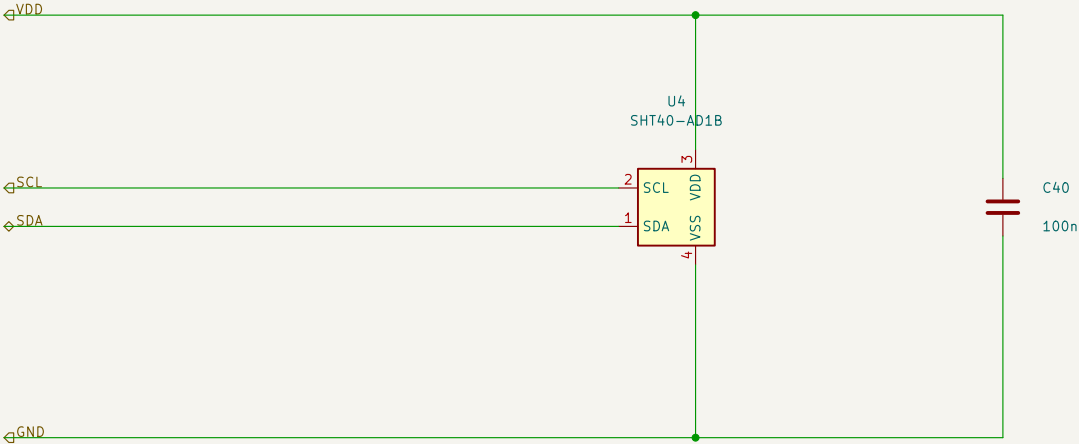
LOGIC SUPPLY | HT7333-1, 30 V input rating



SOT-89 pinout: 1 GND, 2 VIN (tab), 3 VOUT.
Use HT7333-1, not the 12 V HT7333 / 73xx-A part. Confirm exact orderable part before BOM release.
10 uF + 100 nF at input/output per Holtek application circuit; allow for DC-bias derating.
Check LDO dissipation: (VIN - 3.3 V) x ILOGIC. The 250 mA headline is not a thermal allowance.

PulsarFab		
Sheet: /LDO 3.3V/		
File: ldo.kicad_sch		
Title: PulsarDew - 3.3V logic supply		
Size: A4	Date: 2026-10-02	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 4/9

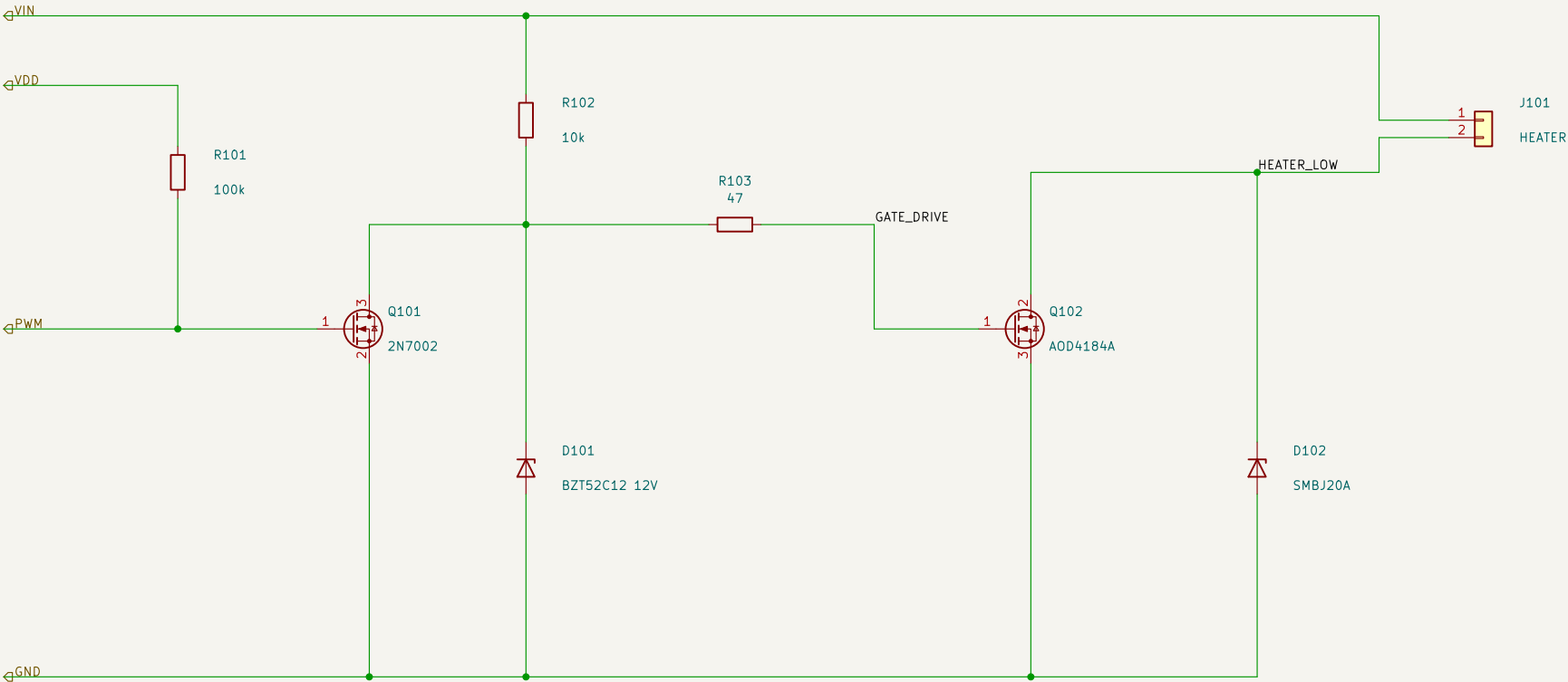
AMBIENT SENSOR | SHT40-AD1B, I2C address 0x44



100 nF at VDD/VSS. I2C pull-ups are on the Controller sheet.
Place the sensor away from the LDO, MOSFETs and heater power copper.

PulsarFab		
Sheet: /SHT40/		
File: sht40.kicad_sch		
Title: PulsarDew – Temperature and humidity		
Size: A4	Date: 2026-10-02	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 5/9

HEATER DRIVER | Inverting, zener-clamped gate drive



PWM HIGH / high impedance with 3V3 present: heater OFF. PWM LOW: heater ON.
Jx01 pin 1 = measured VIN; pin 2 = switched return. Do not bond pin 2 to ground.
Default-off depends on 3V3; loss of logic power can turn the heater ON.
Use slow PWM; verify switching loss, current limit and thermal margins on the bench.

PulsarFab

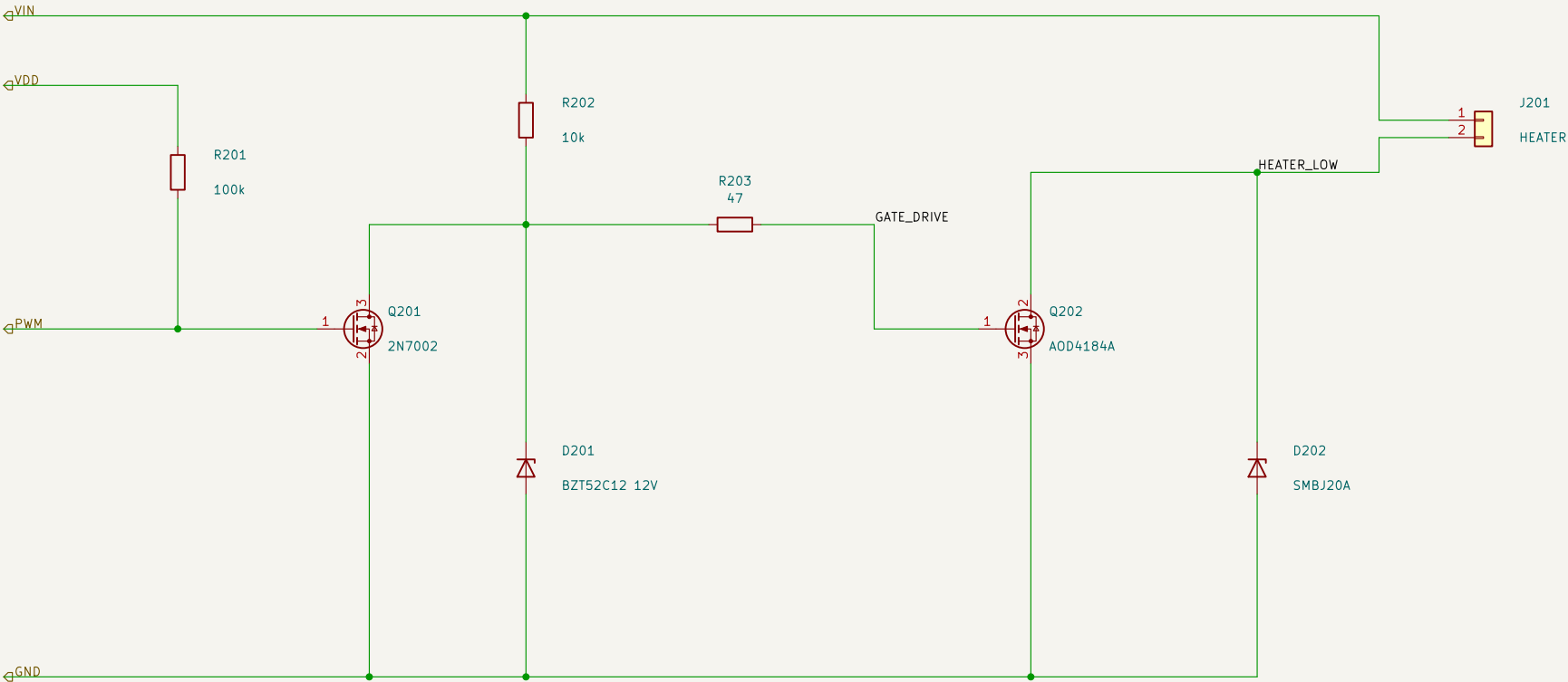
Sheet: /Heater Ch0/
File: heater_channel.kicad_sch

Title: PulsarDew – Heater driver (active-low PWM)

Size: A4 Date: 2026-10-02
KiCad E.D.A. 10.0.6

Rev: 0.2
Id: 6/9

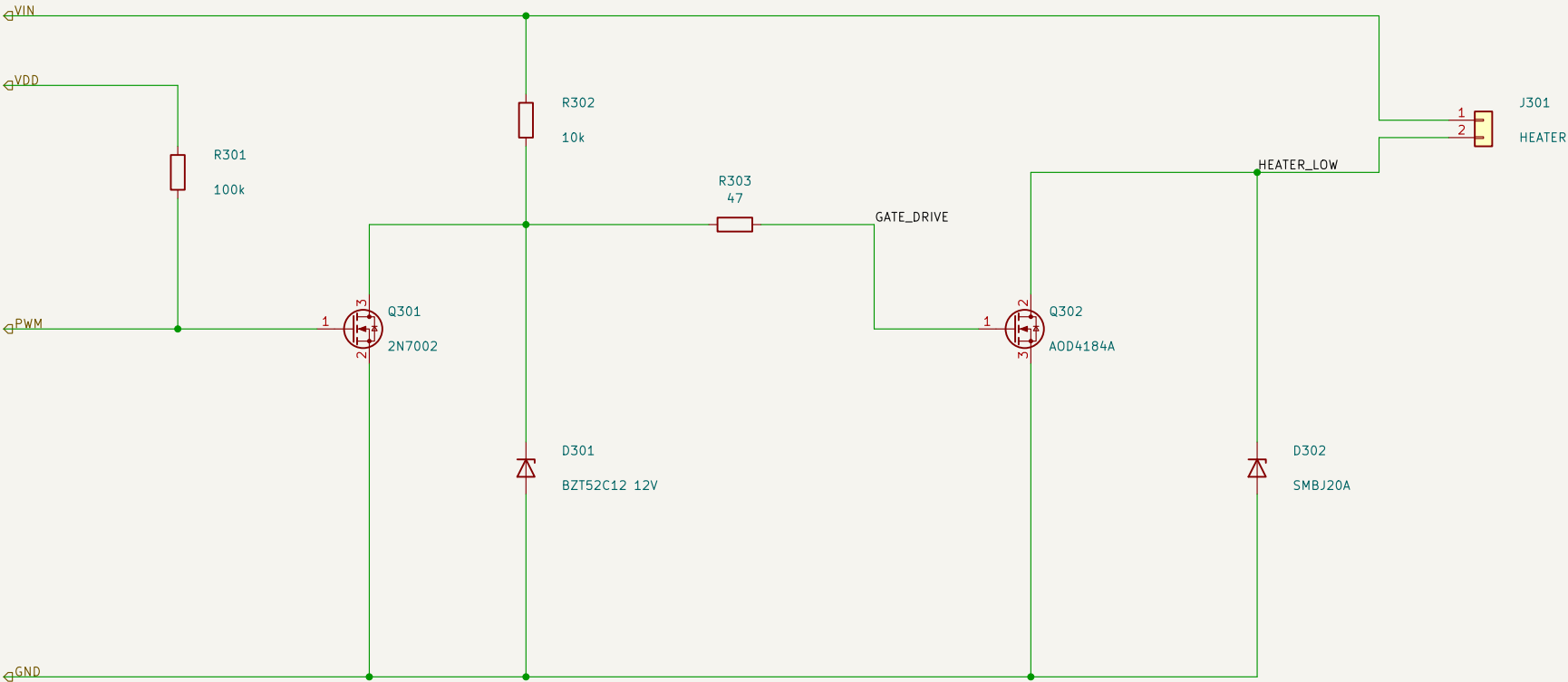
HEATER DRIVER | Inverting, zener-clamped gate drive



PWM HIGH / high impedance with 3V3 present: heater OFF. PWM LOW: heater ON.
Jx01 pin 1 = measured VIN; pin 2 = switched return. Do not bond pin 2 to ground.
Default-off depends on 3V3; loss of logic power can turn the heater ON.
Use slow PWM; verify switching loss, current limit and thermal margins on the bench.

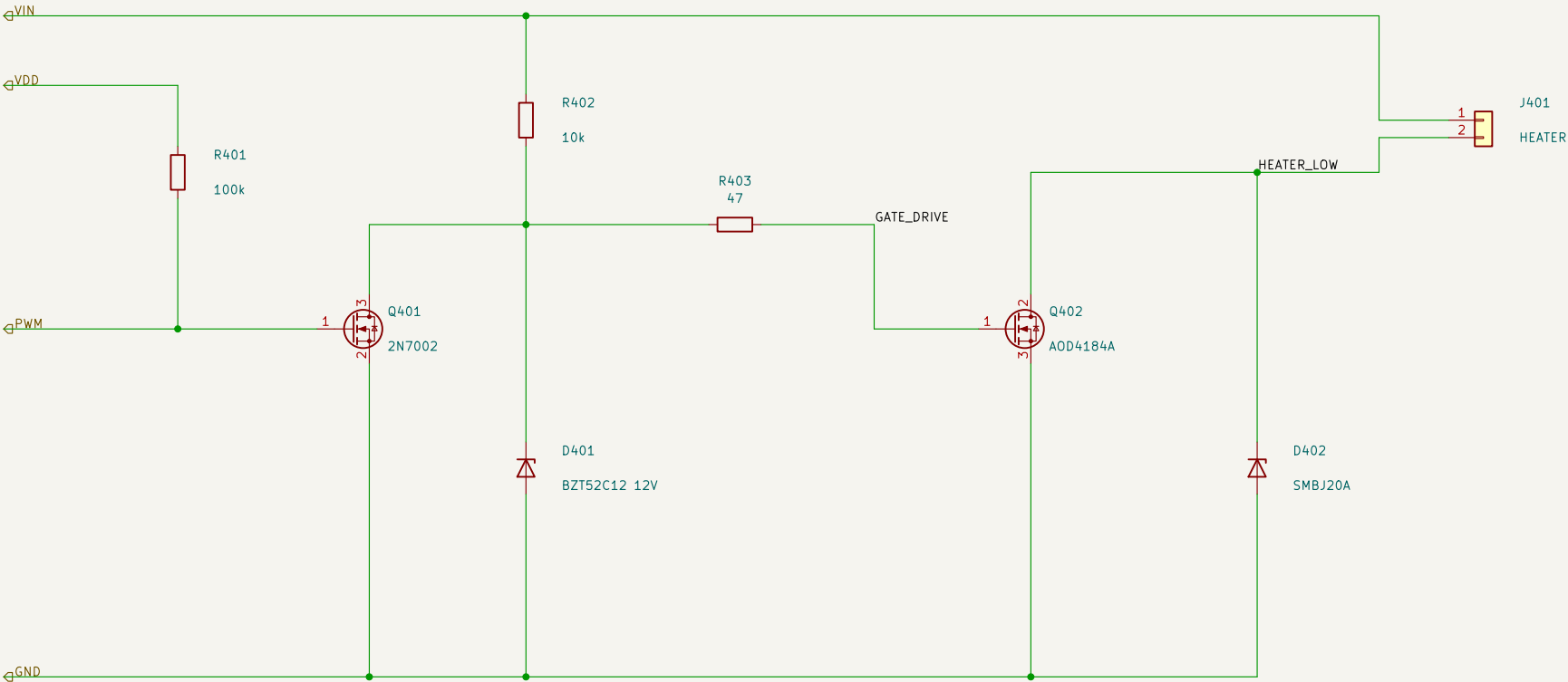
PulsarFab		
Sheet: /Heater Ch1/		
File: heater_channel.kicad_sch		
Title: PulsarDew – Heater driver (active-low PWM)		
Size: A4	Date: 2026-10-02	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 7/9

HEATER DRIVER | Inverting, zener-clamped gate drive



PWM HIGH / high impedance with 3V3 present: heater OFF. PWM LOW: heater ON.
Jx01 pin 1 = measured VIN; pin 2 = switched return. Do not bond pin 2 to ground.
Default-off depends on 3V3; loss of logic power can turn the heater ON.
Use slow PWM; verify switching loss, current limit and thermal margins on the bench.

HEATER DRIVER | Inverting, zener-clamped gate drive



PWM HIGH / high impedance with 3V3 present: heater OFF. PWM LOW: heater ON.
Jx01 pin 1 = measured VIN; pin 2 = switched return. Do not bond pin 2 to ground.
Default-off depends on 3V3; loss of logic power can turn the heater ON.
Use slow PWM; verify switching loss, current limit and thermal margins on the bench.

PulsarFab		
Sheet: /Heater Ch3/		
File: heater_channel.kicad_sch		
Title: PulsarDew – Heater driver (active-low PWM)		
Size: A4	Date: 2026-10-02	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 9/9